Proposal for Data Science Major Applied Concentration

in

Business AI Integration

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MTH 3350 — Data Science Portfolio I — Fall 2025

Carthage College

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Introduction

Within my Data Science major, I've taken 12 courses that have built a strong foundation in data analysis, programming, and overall problem-solving. I've developed experience in R and Python, and some familiarity with C++. Outside of coding, I've learned to apply these skills to real-world data, especially through visualization tools like Tableau, turning data into something useful and easy to understand. My applied concentration builds on this by focusing more on the growing field of Artificial Intelligence, which is inevitable in today's world. It does this by exposing me to AI concepts, teaching me algorithmic practices, and then showing me how to connect them to the world of Economics.

The Data Science Major includes 11 core courses (38 cr.), which are listed below.

- CSC 1030 Data Science I
- CSC 1040 Data Science II
- MTH 1120 Calculus I
- MTH 2040 Linear Algebra
- MTH 3090 Mathematics of Data Science (Spring '27)
- MTH 3350 Data Science Portfolio I

- MTH 4350 Data Science Portfolio II (Fall '26)
- MTH 4990 Senior Thesis Completion (Fall '26)
- MTH 1050 Elementary Statistics
- CSC 1810 Principles of Computer Science I
- BUS 2150 Business Geographics and Data Visualization
- BUS 2110 Business Ethics (Fall '26)

Concentration Course Selection

My concentration focuses on connecting artificial intelligence to real-world applications. It combines coursework in AI, algorithms, and econometrics. Together, these areas help me understand how AI can be used to solve real-world business and economic problems.

- CSC 200F - TP Artificial Intelligence in the Zone
- ECN 3340 - Introduction to Econometrics
- CSC 2560 - Data Structures and Algorithms

The following course descriptions are taken verbatim from the 2025–2026 Carthage College catalog.

- CSC 200F - TP Artificial Intelligence in the Zone
    - This class explores the emerging field of artificial intelligence. It will explore depictions of artificial intelligence in a prominent science fiction television show and code its corresponding artificial intelligence. Students will watch episodes of prominent science fiction television shows including The Twilight Zone that feature robots and artificial intelligence, analyze the major themes of each episode, and write code corresponding to the concepts discussed. This course uses both practical programming and theoretical discussions to truly understand important elements of artificial intelligence design.
- ECN 3340 - Introduction to Econometrics
    - Econometrics is a set of tools researchers use to estimate relationships between variables, test theories, and make forecasts, all using real-world data. Econometric analysis supports decision-making in public policy, business, the court system, and academia. This course provides a rigorous introduction to econometrics, with a particular emphasis on multiple regression analysis. Topics include formulating good research questions; estimating regression models using cross-section, time-series, and panel data; conducting hypothesis tests; and interpreting and critically evaluating published regression results.
- CSC 2560 - Data Structures and Algorithms
    - An examination of advanced programming techniques for problem-solving and manipulating data using primarily object-oriented approaches.

Concentration Rationale

CSC 200F – TP AI in the Zone exposed me to the world of Artificial Intelligence, where we discussed what AI really is and how we can use it. We discussed a wide range of AI topics, from simple chatbots to repetitive jobs that AI could easily replace. I also learned the basics of Python programming. Through both theoretical discussions and coding, I was shown how AI systems are designed and how they function. While we used a classic TV show as an example, the focus was on AI and its logic, which helps build an understanding of how AI can be created and applied to a wide variety of tasks.

CSC 2560 - Data Structures and Algorithms helps me think through problems step by step and understand the most efficient way to solve them. It is important when working with very large datasets or anything related to AI, where inefficient code can really become a problem.

ECN 3340 – Introduction to Econometrics provides the business foundation of my concentration. This course focuses on using real-world data to model relationships, test hypotheses, and make predictions using computer software. Econometrics is directly applicable to business decision-making, connecting data analysis to the economic and business worlds. Econometrics also enables a broader understanding of how the world works. An economic way of thinking encourages analysis of cause and effect, incentives, and trade-offs. These skills are essential not only in academic study but also in professional practice. Economics is a way of understanding how the world works.

When you put these courses together, you get a well-rounded concentration that combines AI concepts, code-based problem-solving, and business data analysis. This combination allows me to understand not only how to build/analyze models, but also how I can apply them to real-world problems and use data to support decision-making.

Conclusion

Overall, my applied concentration in Business AI Integration builds on my Data Science major by applying technical skills to real-world applications. Through coursework in artificial intelligence, data structures/algorithms, and econometrics, I am getting an understanding of how data can be used to solve practical problems and help decision-making.

This concentration also aligns with current industry outlooks, where data science and AI are becoming increasingly important across almost every field. According to the U.S. Bureau of Labor Statistics, employment for data scientists is projected to grow significantly. This is mainly due to the increasing demand for data-driven decision-making in business (and other industries). By combining technical,

analytical, and business courses, this concentration prepares me for future roles/jobs where I can apply AI and data analysis to real-world/business problems.

Beyond data science specifically, AI integration across industries has quickly become the norm. Major areas, including finance, healthcare, retail, and manufacturing, are integrating AI tools into their core operations. From automating routine tasks to providing models. A 2024 McKinsey Global Survey found that a large majority of organizations have integrated AI into at least one business function, and generative AI has seen significant growth in business use cases since 2022. Companies are increasingly looking for professionals who not only understand AI technically but can also understand its outputs and apply them to make decisions.

References

“Computer Science Course Descriptions.” Carthage College, www.carthage.edu/academics/majors-minors/computer-science/course-descriptions/

“Data Science Course Descriptions.” Carthage College, www.carthage.edu/academics/majors-minors/data-science/course-descriptions/

“Economics Course Descriptions.” Carthage College, www.carthage.edu/academics/majors-minors/economics/course-descriptions/

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"The State of AI in 2024." McKinsey & Company, www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai